CV/Resume

Krish Desai, PhD

Employment: Machine Learning Researcher

Lawrence Berkeley National Laboratory

Address: Lawrence Berkeley National Laboratory,

1 Cyclotron Rd, 50-5048,

Berkeley, CA 94720-8099,

United States.

E-mail: krish@desai.ml

Website: https://www.desai.ml

Profiles:

Education

August 2020 – August 2025 University of California, Berkeley

Ph.D., Physics

August 2017 – May 2020 Yale University

M.S., Mathematics

B.S., Mathematics (Intensive)

B.S., Physics (Intensive)

Professional Experience

September 2020 – Present — Lawrence Berkeley National Lab

Machine Learning Researcher

May 2025 – December 2025 — Emissary AI

Machine Learning Engineer

August 2020 – December 2024 — University of California

Associate Instructor

June 2023 – August 2023 — Bridgewater Associates

Investment Analyst Intern

May 2022 – August 2022 — Microsoft Research

PhD Research Intern

May 2020 – July 2020 — Purple Gaze Inc.

Software Developer Intern

Publications

December 2025 — Machine Learning-based Unfolding for Cross Section Measurements in the Presence of Nuisance Parameters

Huanbiao Zhu, **Krish Desai**, Mikael Kuusela, Vinicius Mikuni, Benjamin Nachman, and Larry Wasserman *arXiv:2512.07074*

PDF BibLaTeX

November 2025 Unsupervised Evaluation of Multi-Turn Objective-Driven Interactions

Emi Soroka, Tanmay Chopra, **Krish Desai**, and Sanjay Lall

arXiv:2511.03047

PDF BibLaTeX

October 2025 Unbinned Inference with Correlated Events

Krish Desai, Owen Long, and Benjamin Nachman

European Physical Journal C

PDF BibLaTeX

September 2025 Neural Posterior Unfolding

Fernando Torales Acosta, Jay Chan, **Krish Desai**, Vinicius Mikuni, Benjamin Nachman, Jingjing Pan, Francesco Rubbo

arXiv:2509.06370

PDF BibLaTeX

August 2025 Machine Learning Methods for Cross Section Measurements

Krish Desai

University of California, Berkeley

PhD (Physics) Dissertation

PDF BibLaTeX

December 2024 Neural Posterior Unfolding

Fernando Torales Acosta, Jay Chan, **Krish Desai**, Vinicius Mikuni, Benjamin Nachman, and Jingjing Pan

Thirty-Eighth Annual Conference on Neural Information Processing Systems (NeurIPS), ML4PS Track

PDF BibLaTeX

December 2024 Multidimensional Deconvolution with Profiling

Huanbiao Zhu, **Krish Desai**, Mikael Kuusela, Vinicius Mikuni, Benjamin Nachman, and Larry Wasserman

Thirty-Eighth Annual Conference on Neural Information Processing Systems (NeurIPS), ML4PS Track

PDF BibLaTeX

December 2024 Moment Unfolding: Moment extraction using an unfolding protocol without binning

Krish Desai, Benjamin Nachman, and Jesse Thaler

Physical Review D

PDF BibLaTeX

December 2022 Deconvolving Detector Effects for Distribution Moments

Krish Desai, Benjamin Nachman, and Jesse Thaler

Thirty-Sixth Annual Conference on Neural Information Processing Systems (NeurIPS), ML4PS Track

PDF BibLaTeX

May 2022 SymmetryGAN: Symmetry Discovery with Deep Learning

Krish Desai, Benjamin Nachman, and Jesse Thaler

Physical Review D

PDF BibLaTeX

December 2021 Oblivious points on translation surfaces

Ian Adelstein, **Krish Desai**, Anthony Ji, and Grace Zdeblick

Journal of Geometry

PDF BibLaTeX

December 2021 Symmetry Discovery with Deep Learning

Krish Desai, Benjamin Nachman, and Jesse Thaler

Thirty-Fifth Annual Conference on Neural Information Processing Systems (NeurIPS), ML4PS Track

PDF BibLaTeX

April 2020 Padé Approximants and the Anharmonic Oscillator

Krish Desai

Yale University

BS and MS (Mathematics, Intensive) Thesis

PDF BibLaTeX

Honors & Fellowships

2025 Full Member of Sigma Xi

Sigma Xi

Election to the nation's oldest academic honor society

2020 Howard L. Schultz Prize

Yale University

The most outstanding graduating senior in physics

2020 Distinction in Mathematics

Yale University

2020 Distinction in Physics

Yale University

2020 Lifetime Member of Sigma Pi Sigma

Sigma Pi Sigma

Election to the national physics honor society

2020 George J. Schulz Summer Fellowship

Yale University Chapter

2019 Howard Robert Topol Fellowship

Yale University

2018 James A. Helzer Fellowship

Yale University

2018 SUMRY Fellowship

Yale University

2018 Yale First-Year Research Fellowship

Yale University

2018 Benjamin F. Barge Prize

Yale University

Most outstanding first-year student in mathematics

Talks

December 2024 Multidimensional Deconvolution with Profiling

Thirty-Eighth Annual Conference on Neural Information Processing Systems

Vancouver, Canada

December 2024 Neural Posterior Unfolding

Thirty-Eighth Annual Conference on Neural Information Processing Systems

Vancouver, Canada

July 2024 CMS Seminar

CERN

Geneva, Switzerland

June 2024 Moment extraction using an unfolding protocol without binning

PHY-STAT Unfolding Conference

Paris, France

December 2022 Deconvolving Detector Effects for Distribution Moments

Thirty-Sixth Annual Conference on Neural Information Processing Systems

New Orleans, LA

November 2022 Moment Unfolding with Deep Learning

Machine Learning for Jets

New Brunswick, NJ

June 2022 High Energy Physics Seminar

Korea Institute for Advanced Study

Seoul, South Korea

April 2022 Moment Unfolding using Deep Learning

American Physical Society

New York, NY

December 2021 Symmetry Discovery with Deep Learning

Thirty-Fifth Annual Conference on Neural Information Processing Systems

Virtual

November 2021 Introduction to Symmetry Discovery and Deep Learning

Berkeley Compass Lecture

Berkeley, CA

October 2021 Symmetry Discovery and Deep Learning

Clarifai Perceive Deep Learning AI Conference

Virtual

July 2021 SymmetryGAN

Machine Learning for Jets

Heidelberg, Germany

April 2021 Symmetry Discovery

Lawrence Berkeley National Laboratory, Physics Division Seminar

Virtual

April 2021 Symmetry Discovery using Machine Learning

American Physical Society

Virtual

August 2019 Oblivious Points on Translation Surfaces

Young Mathematicians Conference

Columbus, OH

August 2019 Closed Geodesics on Translation Surfaces

Massachusetts Undergraduate Research Conference

Amherst, MA

Teaching

Fall 2024 PHYSICS 88: Data Science Applications in Physics

Associate Instructor

University of California, Berkeley

Fall 2024 PHYSICS 77: Introduction to Computational Techniques in Physics

Associate Instructor

University of California, Berkeley

Summer 2024 PHYSICS 77: Introduction to Computational Techniques in Physics

Associate Instructor

University of California, Berkeley

Fall 2023 PHYSICS 89: Introduction to Mathematical Physics

Associate Instructor

University of California, Berkeley

Summer 2023 PHYSICS 77: Introduction to Computational Techniques in Physics

Associate Instructor

University of California, Berkeley

Fall 2022 PHYSICS 77: Introduction to Computational Techniques in Physics

Associate Instructor

University of California, Berkeley

Fall 2022 PHYSICS 88: Data Science Applications in Physics

Associate Instructor

University of California, Berkeley

Spring 2022 PHYSICS 8B: Introductory Physics

Head Associate Instructor

University of California, Berkeley

Summer 2021 PHYSICS 7B: Physics for Scientists and Engineers

Associate Instructor

University of California, Berkeley

Spring 2021 PHYSICS 7B: Physics for Scientists and Engineers

Associate Instructor

University of California, Berkeley

Fall 2020 PHYSICS 7B: Physics for Scientists and Engineers

Fall 2020 PHYSICS 7B: Physics for Scientists and Engineers

Associate Instructor

University of California, Berkeley

Leadership & Service

2025 Invited Peer Reviewer

Nature, Scientific Reports

2025 Invited Peer Reviewer

Journal of High Energy Physics

2023-2025 Graduate Mentor

Math and Physical Sciences Scholars, UC Berkeley

2022, 2024, 2025 Invited Peer Reviewer

NeurIPS

2022-2024 Campus Affairs Vice President, and Rules Officer

UC Berkeley Graduate Assembly

2023 Research Mentor

UC Berkeley Physics Innovators Program

2021 – 2024 Committee Member

UC Berkeley Physics Faculty Search

2018-2019 Vice President

Skills

Programming Languages	Python	Julia	C
Machine Learning	TensorFlow/Keras	PyTorch	scikit-learn
	Deep Learning	Bayesian Methods	Generative Models
Data Science	Statistical Inference	Bayesian Statistics	Monte Carlo
	Unfolding	Deconvolution	
Physics & Mathematics	Particle Physics	Quantum Field Theory	Stochastic Calculus
	Operator Theory	Differential Geometry	
Computing	HPC	Git	Docker
Languages	English	French	Hindi
	Gujarati		